

Enrolment No:.....

Name of Student.....



BENNETT
UNIVERSITY
THE TIMES GROUP

NAAC
GRADE A+
ACCREDITED UNIVERSITY

End Semester Examinations, May 2026
Even Semester 2025-26
B. Tech.(CSE) - IVth Semester
(CSET-244) Design and Analysis of Algorithms

Time: 2hrs.

Max Marks: 40

Note: Student must attempt 6 Questions in total. First question is compulsory and student must attempt question(s) from each section. Marks are indicated against each question.

PART-A

Q.1 First question will consist of 8 small questions covering entire syllabus out of which student has to attempt 6 questions.

- a. Define the algorithm and its various characteristics in detail. [CO-1] [L-1]
b. Determine recurrence relation for the following code: [CO-1] [L-3]
void complexRecursive(int n) {
if (n < 1) return;
doConstantWork(); // O(1)
complexRecursive(n/3);
complexRecursive(n/3);
complexRecursive(n/3); // Three calls on 1/3rd size }
c. Distinguish between the Greedy approach and Dynamic Programming by [CO-2] [L-2]
examining how each utilizes the Principle of Optimality.
d. Given two 2x2 matrices: [CO-2] [L-3]

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

The submatrices are defined as:

$$A_{11} = 1, A_{12} = 2, A_{21} = 3, A_{22} = 4$$

$$B_{11} = 5, B_{12} = 6, B_{21} = 7, B_{22} = 8$$

Using Strassen's Matrix Multiplication method, Compute the seven products M_1 to M_7 .

- e. Demonstrate the role of a State-Space Search Tree in solving the Four Queen's [CO-3] [L-3]
problem.
f. Differentiate between Hamiltonian Cycle and Travelling Salesman Problem by [CO-3] [L-2]
listing any two key differences.
g. Describe the decision version of the Traveling Salesman Problem and identify [CO-4] [L-2]
which complexity class it belongs to.
h. Define the primary difference between the complexity classes P and NP. [CO-4] [L-1]

PART-B

Attempt any 2 questions, Each question carry 6 marks.

6*2=12

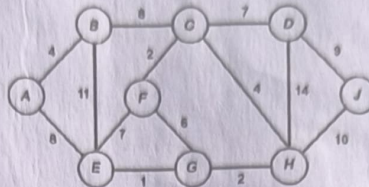
- Q.2.** A file comparison tool is designed to identify common patterns between two versions of a [CO-2] [L-3]
document to highlight similarities. Given two strings "AGGTAB" and "GXTXAYB"

representing two document versions, determine the longest sequence of characters that appears in both (not necessarily contiguously). Use an appropriate dynamic programming approach and show all steps.

- Q.3. Apply the MERGE procedure of Merge Sort on the array: $\langle 10, 5, 4, 15, 3, 20, 8 \rangle$ and show all steps. [CO-2] [L-3]6

- Q.4. Consider a weighted complete graph

[CO-2] [L-3]6



Apply Prim's Algorithm to the given weighted graph to construct a minimum spanning tree. Show all intermediate steps and compute the total weight of the tree.

PART-C

Attempt all questions, Each question carry 8 marks.

8*2=16

- Q.5. A scheduling system must select a maximum number of non-overlapping activities from a set of competing tasks with given start and finish times. [CO-2] [L-3]8

(a) Apply an appropriate greedy strategy to select the optimal set of activities from the following data. Show all steps.

Jobs: {A1, A2, A3, A4, A5, A6, A7, A8, A9, A10}

Start Times (S_i): {1, 3, 0, 5, 8, 5, 6, 8, 12, 2}

Finish Times (F_i): {4, 5, 6, 7, 9, 9, 10, 11, 14, 13}

(b) Evaluate why the greedy strategy works for this problem and discuss whether this approach would still be valid if the objective were to maximize total duration instead of the number of activities.

OR

A logistics startup based in Bengaluru needs to optimize its delivery route across four major tech hubs. The distances (in km) between the hubs are provided in the adjacency matrix below. Find the shortest possible route that visits each city exactly once and returns to Bengaluru. Use the Branch and Bound algorithm. Show the state-space tree and explain the lower bound calculation at each step. Note: Using a Brute Force approach will result in a 50% marks penalty. [CO-2] [L-3]8

Cities	Bengaluru	Hyderabad	Chennai	Mysuru
Bengaluru	∞	10	15	20
Hyderabad	5	∞	9	10
Chennai	6	13	∞	12
Mysuru	8	8	9	∞

- Q.6. A text processing system needs to efficiently search for patterns in large documents where repeated comparisons are costly. [CO-3] [L-3]8

(a) Design an efficient string matching approach that avoids redundant comparisons by preprocessing the pattern. Apply your approach to find all occurrences of the pattern "xyxy" in the text "xyxyxyxyxy". Show all steps.

(b) Compare your approach with the naive string-matching method in terms of efficiency.

OR

A software system needs to perform a sequence of matrix multiplications for efficient data processing. The dimensions of the matrices are given as (20, 5, 35, 10, 15, 30, 6). [CO-3] [L-3]8

(a) Determine the optimal order of multiplication that minimizes the total number of scalar multiplications using dynamic programming. Show all steps.

(b) Analyze how the choice of parenthesizing affects computational cost and explain why the obtained solution is optimal.

xyxy